

Validity Analysis: AFRIDOC MT

Target context: Amhara Region Agricultural & Microfinance Document Translation — Cooperative and Rural MFI End-Users

Overall risk: **HIGH**

Deployment Context

Use case: Document-level translation of agricultural input subsidy notices, land-use policy updates, and credit-scheme terms from federal and regional bureaus into Amharic

Target population: Farmer cooperative leaders and rural microfinance clients (e.g., ACSI, ADCSI borrowers) in Amhara interpreting government and lender document

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology $\triangle$	2	Significant gaps	high
Input Content $\triangle$	1	Serious concern	high
Input Form $\checkmark$	4	Minor gaps	high
Output Ontology $\triangle$	2	Significant gaps	high
Output Content $\triangle$	2	Significant gaps	high
Output Form $\checkmark$	3	Moderate gaps	medium
Average	2.3		

$\triangle$ = highest concern $\checkmark$ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

AFRIDOC-MT is a high-quality document-level MT benchmark for five African languages including Amharic in Ethiopic script, but its fitness for evaluating Amharic agricultural and microfinance document translation in Amhara region is severely limited by three convergent gaps: (1) total domain absence — no agricultural, land-tenure, cooperative governance, or credit-scheme content exists in the benchmark, confirmed empirically across all sampled examples and corroborated by web research showing no other Amharic MT benchmark covers these domains either; (2) source-language direction mismatch — the benchmark translates from English to Amharic and exhibits acknowledged translationese, whereas the deployment operates on documents natively authored in Amharic proclamation register; and (3) annotator-authority mismatch — the four named

Amharic translators were prepared for health/IT domains with no documented agricultural or legal Amharic expertise, while the user-specified preferred ground-truth authority is federal bureau translators. Output scoring is calibrated for narrative discourse coherence with no metric for terminological consistency — the critical coherence requirement for legal-administrative documents. The benchmark's strongest alignment is on Input Form (text-only Amharic in Ethiopic script with documented tokenization accommodations) and Output Form (text output modality match). Of the three HIGH-priority dimensions (IO, IC, OC), all score 1-2.

Practical Guidance

What This Benchmark Measures

AFRIDOC-MT can credibly measure: (a) general document-level MT capability for English-to-Amharic in Ethiopic script using NLLB-200, GPT-4o, and other current systems [Q4, Q41]; (b) basic discourse coherence in narrative health and IT news [Q16, Q55]; (c) fluency-vs-accuracy tradeoffs for Amharic at sentence-level vs. pseudo-document granularities [Q61, Q72]; (d) which model architectures struggle with non-Latin script and morphological richness [Q53, Q127]. The strongest dimensions for the deployment are Input Form (4) and Output Form (3), reflecting that the benchmark technically handles Amharic text in Ethiopic script and produces text outputs at relevant granularities.

Construct Depth

Construct depth is shallow for the deployment's specific construct (legal-administrative Amharic translation quality). The benchmark probes general document-level MT depth in news domains via a multi-evaluation stack (d-chrF + d-BLEU + GPT-4o-judge + human DA), but the GPT-4o judge is explicitly unreliable for African languages [Q67, Q85, Q219], leaving human DA with Amharic Krippendorff's alpha of only 0.46 [Q71] as the residual signal. No measurement instrument exists for terminological consistency across clauses [WEB-25, WEB-26 confirm metric gap], structural fidelity for numbered clauses, register-appropriateness against Ethiopian proclamation conventions, or accessibility for variable-literacy end-users. Performance on AFRIDOC-MT does not predict performance on deployment-domain documents.

What Else You Need

Critical supplementation: (1) construct a deployment-domain test set drawing from publicly available Ethiopian proclamation Amharic (e.g., Proclamation 985/2016 [WEB-10], Proclamation 456/2005 [WEB-12]) and ACSI/EABC notices [WEB-3, WEB-5] — closing the IO and IC gaps; (2) recruit federal Ministry of Agriculture or Amhara Regional Bureau translators as annotators for that test set — closing the OC gap; (3) implement a terminology-consistency metric adapted from Semenov-Bojar or HHI [WEB-25, WEB-26] for proclamation-derived term categories — closing the OO gap; (4) add structural-fidelity and plain-language accessibility scoring if format-preserving or oral-mediation use cases are confirmed — closing the OF gap; (5) collect Amharic-source-language examples to address the source-direction mismatch.

ID	Evidence
Q4	"Our results indicate that NLLB-200 achieves the best average performance among the standard NMT models, while GPT-4o outperforms general-purpose LLMs." (p.1)
Q16	"Both document-level and context-aware MT allow for the possibility of improving translation quality for context-dependent phenomena such as coreference resolution (Müller et al., 2018; Bawden et al., 2018; Voita et al., 2018; Herold and Ney, 2023), lexical disambiguation (Rios Gonzales et al., 2017; Martínez Garcia et al., 2019), and lexical cohesion (Wong and Kit, 2012; Garcia et al., 2014, 2017; Bawden et al., 2018; Voita et al., 2019)." (p.2)
Q41	"We present the result of 12 models in total, including the 1.2B version of Toucan, 1.3B and 3.3B versions of NLLB-200, 3B and 13B versions of MADLAD-400 and Aya-101 respectively. We also have the 8B instruction tuned version of LLama3.1 (LLama3.1-IT), 9B version of Gemma-2 (Gemma2-IT), and LLaMAX3-Alpaca." (p.4)
Q53	"We report d-chrF scores for the best prompt per model and language direction in the main text, as chrF better captures the morphological richness of African languages (Adelani et al., 2022), with full results provided in Appendix C." (p.5)
Q55	"Following Sun et al. (2025), we assess each translated document's fluency, content errors (CE), and cohesion errors—specifically lexical (LE) and grammatical (GE) errors—using GPT-4o, with evaluation limited to a few model outputs due to cost constraints (Appendix B.6)." (p.5)

Q61	“Pseudo-document translation is worse than sentence-level translation when translating into African languages Our results from pseudo-document translation show a performance drop across different models compared to sentence-level translation, especially when translating into African languages.” (p.6)
Q67	“These inconsistencies raise concerns about GPT-4o's reliability.” (p.7)
Q71	“We obtained Krippendorff's alpha values of ≥ 0.40 , which are relatively low due to the fine granularity of the evaluation scale.” (p.8)
Q72	“Human evaluation results align closely with d-chrF, which favors sentence-level translations over pseudo-document translations when translating into African languages.” (p.8)
Q85	“However, our DA and qualitative analysis found GPT-4o's judgments inconsistent for African languages, and sentence-by-sentence translation proved more effective for some languages.” (p.9)
Q127	“Our analysis revealed that Amharic and Yorùbá—languages with unique characteristics such as non-Latin scripts and diacritics, respectively—had the largest average token counts across the tokenizers.” (p.19)
Q219	“These results suggest that using GPT-4o as a translation judge is not yet well-suited for low-resource languages.” (p.26)
WEB-3	EABC and primary cooperative unions are the institutional terminology anchors for deployment, distinct from health/IT translator preparation (https://addisstandard.com/sowing-under-shadows-of-violence-amhara-farmers-face-bleak-harvest-as-conflict-deepens-fertilizer-shortages-drives-up-input-prices/)
WEB-5	ACSI serves ~2 million clients across all woredas; key MFI in deployment (https://www.findevgateway.org/sites/default/files/publications/files/mfg-en-case-study-a-review-of-ethiopian-microfinance-institutions-and-their-role-in-poverty-reduction-a-case-study-on-amhara-credit-and-saving-institution-acsi-jan-2011.pdf)
WEB-10	Ministry of Justice provides authoritative Amharic terminology — federal bureau translators are the preferred authority unrepresented in the benchmark (https://justice.gov.et/en/law/cooperative-societies-proclamation-2/)
WEB-12	Land Administration Proclamation 456/2005 defines land-tenure terminology absent from benchmark (https://land.igad.int/index.php/countries/37-countries/ethiopia/38-ethiopia-profile?start=2)
WEB-25	Semenov & Bojar (WMT 2022) terminology-consistency metric exists but not validated for African languages (https://aclanthology.org/2022.wmt-1.41/)
WEB-26	HHI applied to legal corpora terminology consistency — not validated for Amharic (https://www.researchgate.net/publication/358368108_Measuring_Terminology_Consistency_in_Translated_Corpora_Implementation_of_the_Herfindahl-Hirshman_Index)